

Unit-4

Multimodal Applications

MULTIMODAL MACHINE LEARNING

MULTIMODAL AI

Bridging Language, Vision & Sound



The Modality Spectrum

- **Text:** Discrete tokens, semantic richness (NLP).
- **Audio:** Waveforms, spectrograms, prosody (Speech).
- **Image/Video:** Spatial and temporal pixel correlations.
- **Sensors:** LiDAR, IMU, GPS, Heat maps (Robotics/IoT).

Representation & Alignment

- **Joint Representations:** Mapping all modalities into a single shared vector space.
- **Coordinated Representations:** Keeping modalities separate but enforcing similarity (e.g., Contrastive Learning).
- **Alignment:** Finding the direct correspondence between sub-elements (e.g., which word matches which video frame?).

Representation & Alignment

- Alignment is perhaps the hardest part of multimodal AI. It is categorized into two types:
 - **Explicit Alignment:** Directly matching parts of the data. For example, in **Visual Grounding**, we align the phrase "the red ball" with the specific bounding box in an image.
 - **Implicit Alignment:** The model learns to attend to relevant parts of one modality while processing another (e.g., **Attention Mechanisms** in Transformers where the text query "looks" at specific image patches).

Co-learning & Modality Relationships

- **Co-learning:** Transferring knowledge from one modality to help another.
- **Complementary:** Modalities provide different info (e.g., Text says "happy," Video shows a smile).
- **Redundant:** Modalities provide the same info (e.g., Audio "Stop" and a red light).

Co-learning (The "Teacher-Student" Paradigm)

- Co-learning allows a model to learn a task in one modality by leveraging the labels or features of another.
- **Example:** You have a massive dataset of videos with audio, but only a small dataset of just audio. You can use the visual information to help the model learn better audio features (cross-modal transfer).

Complementary vs. Redundant

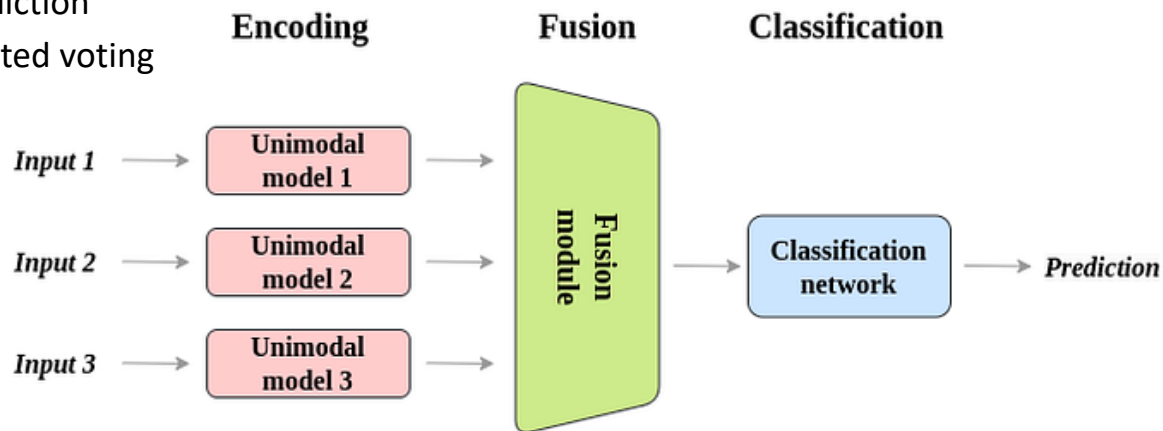
- **Redundancy** increases **reliability**. If the microphone is noisy, the visual lip-reading (redundant signal) helps the model stay accurate.
- **Complementarity** increases **context**. A video of a person crying could be "sadness" or "joy." The audio (sobbing vs. laughing) provides the complementary info needed to disambiguate.

The "Nightmare" Scenarios (Challenges)

- Missing Modality (The "Broken Sensor" problem).
- Modality Imbalance (One signal dominates the others).
- Temporal Alignment (The "Lag" problem).
- Noisy Modalities (Low-quality or corrupted inputs).

Model-Free Fusion Approaches

- Early Fusion
 - Image_features + Audio_features → Neural Network
- Intermediate Fusion
 - Image model → Embedding
 - Audio model → Embedding
 - Joint embedding → Neural Network
- Late Fusion
 - Image model → prediction
 - Audio model → prediction
 - Final output = weighted voting



Hybrid Fusion

- **1. Hierarchical (Multi-Stage) Hybrid Fusion**

- This is the most common form. It performs fusion at multiple levels of the neural network.
- **The Workflow:** Local features from two modalities (e.g., Audio and Video) are fused early on to create "mid-level" features. These mid-level features are then processed alongside a third modality (e.g., Text) at a later stage.

- **2. Parallel (Joint-Decision) Hybrid Fusion**

- In this setup, the model runs an Early Fusion branch and a Late Fusion branch simultaneously.
- The Workflow: 1. Branch A: Concatenates features at the input level to learn joint correlations. 2. Branch B: Processes modalities independently to maintain the integrity of each signal. 3. The Hybrid Step: The outputs of Branch A and Branch B are merged (usually via a weighted average or a meta-classifier) to make the final prediction.

- **3. Gated or Adaptive Hybrid Fusion**

- This is a more "intelligent" version of hybrid fusion that uses an internal mechanism to decide **when** and **how much** to fuse.
- **The Workflow:** It uses **Gated Neural Networks** (similar to LSTMs or GRUs). A "Controller" or "Attention" sub-network looks at the input quality. If the Audio is noisy, the gate suppresses the early-fusion features and relies more on the Late-fusion visual features.

Kernel Based Fusion

- Most multimodal data is not linearly separable in its raw form. For example, you cannot draw a straight line between "Happy" and "Sad" based on raw pixel values and audio frequencies.
- **The Math:** A kernel function $k(x, y)$ calculates the inner product of two data points in a high-dimensional feature space without ever explicitly calculating the coordinates in that space.
- **Multimodal Application:** We define a specific kernel for each modality:
 - K visual for images (e.g., an RBF kernel).
 - K audio for sound (e.g., a Polynomial kernel).
 - K text for text (e.g., a String kernel).

Multiple Kernel Learning (MKL)

- MKL is the advanced version of kernel fusion. Instead of picking one "perfect" kernel, we use a **weighted combination** of several kernels.
- **The Fusion Equation**
 - The fused kernel K_{fused} is defined as:

$$K_{fused}(x, y) = \sum_{m=1}^M \beta_m K_m(x_m, y_m)$$

Types

- **A. Early Kernel Fusion (Composite Kernels)**
 - We combine the data *inside* the kernel function.
 - **Pro:** Captures cross-modal correlations.
 - **Con:** Requires all modalities to be present and synchronized.
- **B. Intermediate Kernel Fusion (MKL)**
 - This is the most popular type. We compute separate kernel matrices for each modality and then combine the matrices.
 - **Pro:** You can use different types of kernels for different data (e.g., a Histogram Intersection kernel for images and a Linear kernel for text).
- **C. Late Kernel Fusion**
 - We train separate SVMs (Support Vector Machines) for each modality kernel and fuse their output scores (similar to Late Fusion in Neural Networks).

Multimodal Graphical Models

- Multimodal learning studies how heterogeneous, complex descriptors can be optimized to create a robust, generalizable learning system
- Challenges exist such as modality collapse or missing modalities
- Different modalities lead to intricate relational dependencies that graph learning can leverage by building and learning on optimally defined graphs

Why graphical models?

- In many multimodal problems, the data is sequential (video frames, speech over time, gesture movements). We need models that capture how different modalities evolve TOGETHER over time.

HMM

- A Hidden Markov Model has ONE chain of hidden states that evolves over time, and at each time step, it produces ONE observation. Example: speech recognition — hidden states are phonemes, observations are audio features.
- **Problem with regular HMM:** What if you have TWO modalities (audio + lip movement) that BOTH evolve over time with their OWN dynamics? A single HMM chain cannot capture this.

FHMM

Regular HMM (single modality):

Hidden: $s1 \rightarrow s2 \rightarrow s3 \rightarrow s4$ (one chain)
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
Observed: $o1 \quad o2 \quad o3 \quad o4$ (audio features)

Factorial HMM (two modalities, two PARALLEL chains):

Audio hidden: $a1 \rightarrow a2 \rightarrow a3 \rightarrow a4$ (audio state chain)
Visual hidden: $v1 \rightarrow v2 \rightarrow v3 \rightarrow v4$ (visual state chain)
 $\searrow \downarrow \swarrow \quad \searrow \downarrow \swarrow$
Combined observation: $o1 \quad o2 \quad o3 \quad o4$

Each observation depends on BOTH the audio state AND visual state.
The two chains evolve independently but jointly explain the observation.

Factorial Hidden Markov Models

- Imagine you are monitoring a person's health using a **Biometric System** (one of your case studies). You have:
- **Chain 1 (Hidden)**: The person's actual physical exertion level (Resting, Walking, Running).
- **Chain 2 (Hidden)**: The person's emotional state (Calm, Stressed, Excited).
- Both of these "hidden" chains exist simultaneously. You don't see them directly. Instead, you observe **one combined signal**: a heart rate monitor (Observation).

Mathematical Structure

2. Mathematical Structure

In a standard HMM, we have one state q_t . In an FHMM with M layers, the state at time t is a collection of M states:

$$S_t = \{s_t^{(1)}, s_t^{(2)}, \dots, s_t^{(M)}\}$$

The Three Key Components:

1. **Independent Transitions:** Each chain m evolves independently according to its own transition matrix $P(s_t^{(m)} | s_{t-1}^{(m)})$.
2. **Joint Observation:** The observation O_t depends on the current state of **all** chains:
 $P(O_t | s_t^{(1)}, \dots, s_t^{(M)})$.
3. **Factorization:** The "Factorial" name comes from the fact that the state space is the Cartesian product of the individual chains' state spaces.

Why it's Crucial for Multimodal Fusion

- FHMMs are the gold standard for **Source Separation** (The "Cocktail Party Problem").
- **Example:** You have one microphone (Observation) recording two people talking (Two Hidden Chains).
- **The Problem:** A standard HMM would need $N \times M$ states to model this, which leads to "State Explosion."
- **The FHMM Solution:** It models them as two separate processes that happen to be heard by one sensor.

CRF

- A Conditional Random Field (CRF) is like an HMM but discriminative — it directly models the probability of the label sequence given the observations, without modeling the observation distribution. This makes it more flexible for classification.

Multi-view Hidden CRF

Each 'view' is a different modality. The model has hidden states that capture the joint dynamics across views. The key advantage over FHMM: CRFs can use overlapping, arbitrary features from each view, and they directly optimize for the classification objective.

Multi-View Hidden CRF:

View 1 (video): visual features at each time step

View 2 (audio): audio features at each time step

View 3 (text): transcript features at each time step

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Hidden states: $h_1 \rightarrow h_2 \rightarrow h_3 \rightarrow h_4$ (shared hidden chain)

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Labels: $y_1 \ y_2 \ y_3 \ y_4$ (e.g., emotion labels)

All views feed INTO the hidden states.

The model learns which view to trust at each time step.

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Multi-view Hidden CRF

- **Application:** Emotion recognition in videos. At some moments, the face (visual) is most informative. At other moments, the voice tone (audio) reveals more. Multi-view CRF dynamically weights each modality at each time step.
- **Modern perspective:** Factorial HMMs and multi-view CRFs have been largely replaced by Transformer-based models with cross-attention (like BLIP). However, they are still used in small-data sequential problems and are important for understanding the foundations of multimodal temporal modeling.

End of Slide

THANK YOU

